

Marine Microbial Biodiversity

Lecture 00

Matteo Selci, PhD

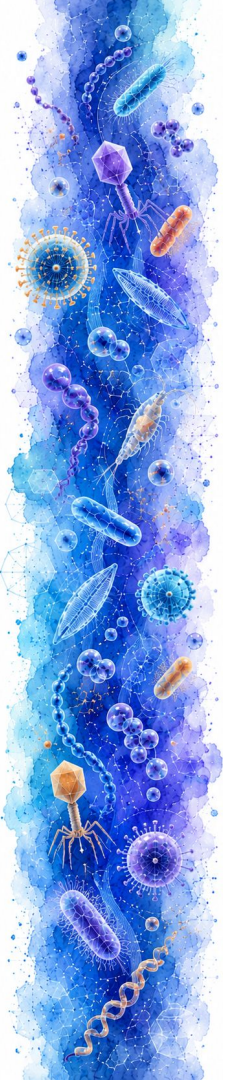
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📷 @matteo.selci

🐦 @matteo_selci





My Background

2016 — B.Sc. (L) in Biological Science, University of Roma Tre

2019 — M.Sc. (LS) in Marine Biology, Polytechnic University of Marche

2019 — Visiting Scholar, OVSICORI-UNA, Costa Rica

2023 — PhD in Biology, University of Naples “Federico II”

2025 — Postdoc, Rutgers University, USA

Present — Postdoc, University of Naples “Federico II”

Current Position

PostDoc Researcher

Department of Biology, University of Naples "Federico II"



Research subject

Ecology, diversity, and metabolic potential of subsurface microbial communities associated with natural hydrogen-rich systems and underground hydrogen storage reservoirs

Hy^{DRA}



**9 Research
Partners**



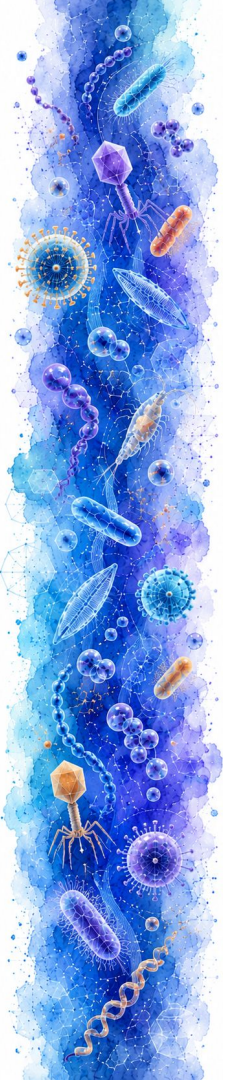
6 Countries



**10+ Storage Site
Operators**



30+ Researchers



My Approach

I am fundamentally interested in the processes that link the biosphere and geosphere, with a focus on the microbial world and how diversity and function are expressed in extreme environments

My research explores:

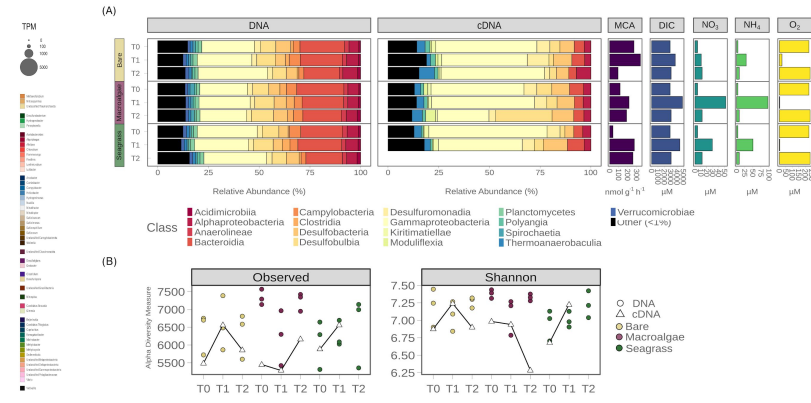
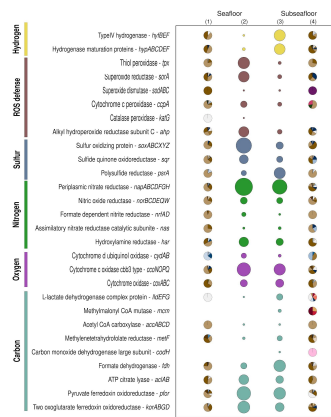
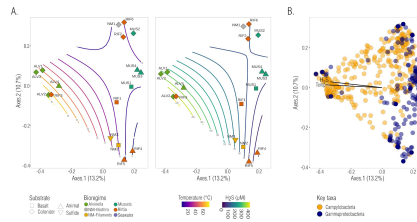
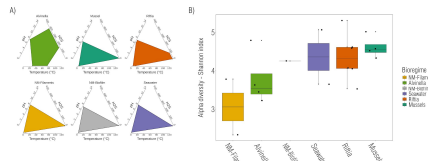
1. Ecosystem functioning by integrating multi-scale biological data (e.g., from genes to ecosystems)
2. Examining interactions between biotic and abiotic components, including rocks, sediments, and geochemical gradients

How to:

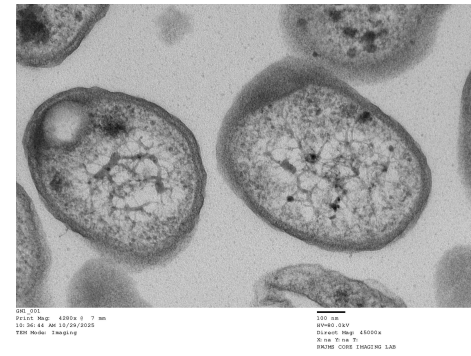
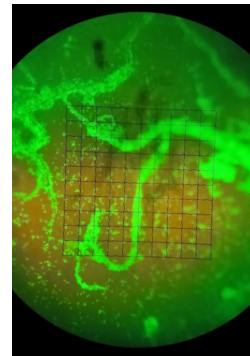
Combining environmental surveys, comparative genomics, and computational approaches to investigate how microbial life adapts and thrives across diverse chemosynthetic ecosystems

My Background

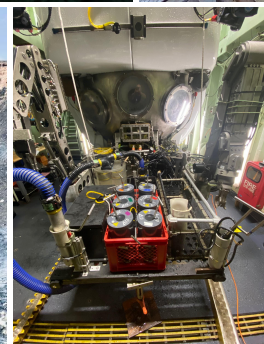
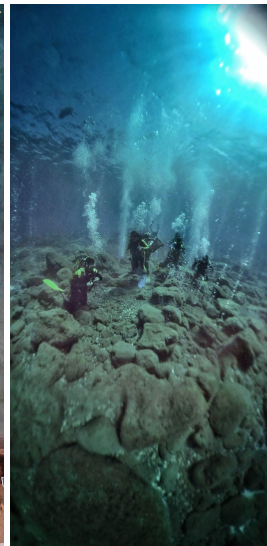
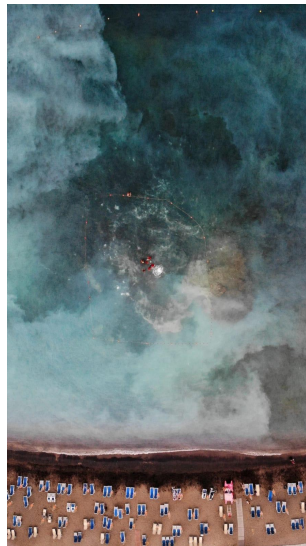
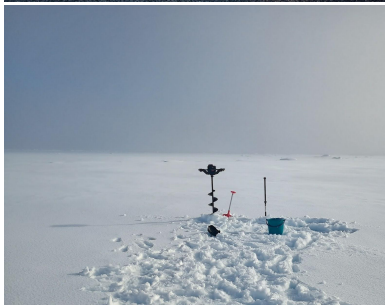
Culture independent

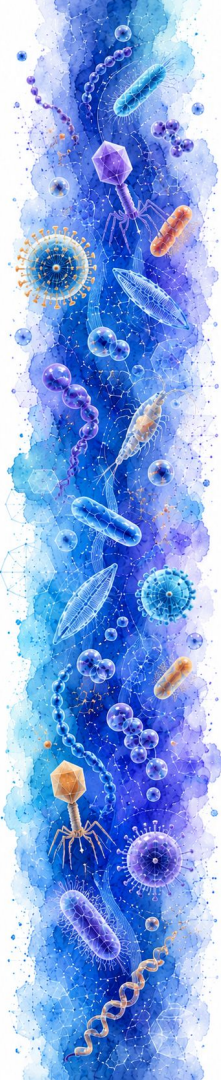


Culture dependent



Best Job I Ever Had!

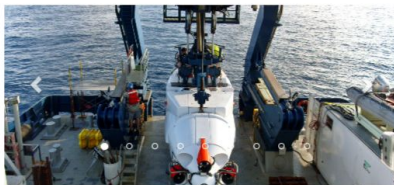




Welcome to the Giovannelli Lab

Hi, my name is Donato Giovannelli and I am a Professor of Microbiology at the Department of Biology of the University of Naples "Federico II", in Italy. My group aims to explore and understand the emergence and evolution of life and the co-evolution of the Geosphere and the Biosphere. I've been recently awarded a **ERC Starting Grant** to look at the co-evolution of biogeochemically-relevant proteins and trace metal availability in the environment called CoEvolve. Read more about the [CoEvolve project](http://www.coevolve.eu) here.

www.coevolve.eu



In my lab we combine classic microbiology techniques with data from comparative genomic, phylogenetic, geochemistry, environmental surveys and computational approaches to reconstruct geo-bio interactions. The majority of our work is carried out in **extreme environments**, ranging from deep-sea and shallow-water hydrothermal vents, hot springs and volcanoes, the subsurface and Polar regions.

Our laboratory logo (in its version 2.0), an abstract representation of what we call "microbiology in planetary context". Designed by Patricia Barcala Dominguez



News

11. September 2024

New postdoctoral fellowship position in bioinformatic available in my group! [Read more here!](#)

31. Oct 2023

The podcast ABISSI: Diario dal Fondali del Pacifico! made with Rai RadioScienze and RaiPlay Sound is finally out! You can listen to the 8 episodes on [RaiPlay Sound](#)

19. Nov 2022

The scientific drilling in the framework of the international project ICDP DIVE is on-going! Current depth is 400 meters and counting...

12. Oct 2021

Donato was guest of the radio program SCARTI RADIOATTIVI (puntata 4 - in Italian) "Cronache dalla Terra di Sotto" organized by the ektemporanea group within FestivalLetteratura di Mantova.

4. Jul 2021

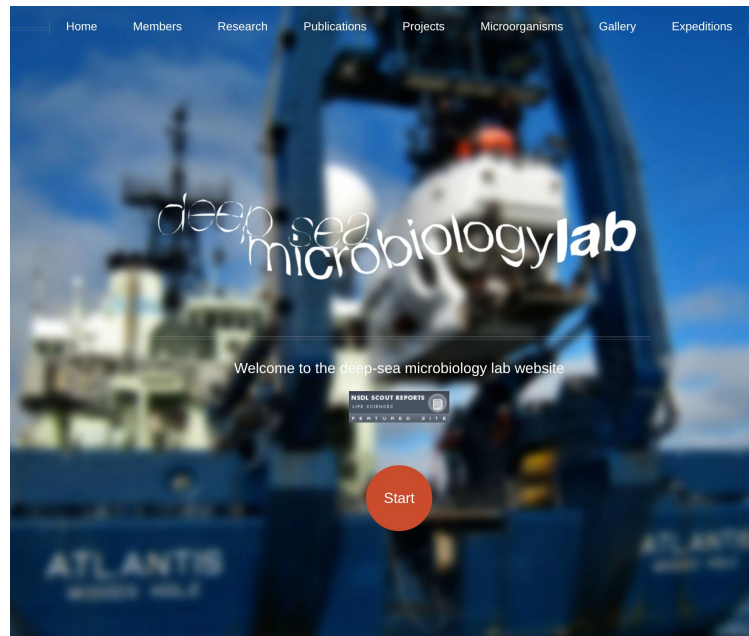
A short video interview (in Italian) recorded while I was in Iceland sampling the shallow water vents of Strytan appeared today on the Swiss Radio Television Website. You can see the fantastic full 4 minute interview on the [RSI website](#)

... see all News

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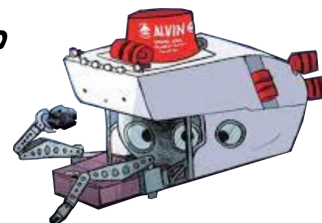
www.coevolve.eu

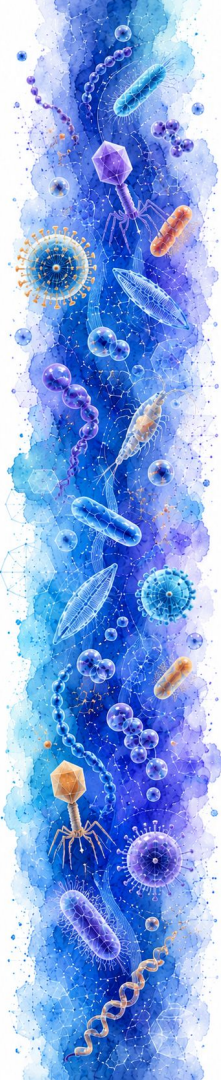
 GIOVANNELLILAB



www.deepseamicrobiologylab.com

deep sea microbiologylab





COURSE STRUCTURE

DURATION: 24 FRONTAL LESSONS (48 HOURS)

RELEVANT RESEARCH PAPER TO READ AND DISCUSS

INVITED GUEST LECTURERS

THE EXAM WILL CONSIST OF:

2 MULTIPLE-CHOICE TESTS -> 30 QUESTIONS IN 30 MINUTES

GROUP PRESENTATIONS (CONTRIBUTING UP TO 3 POINTS): **14 DEC 2026**

FINAL GRADE: STUDENTS MAY EITHER CONFIRM THEIR ACCUMULATED FINAL SCORE OR CHOOSE TO SIT FOR AN ORAL EXAMINATION

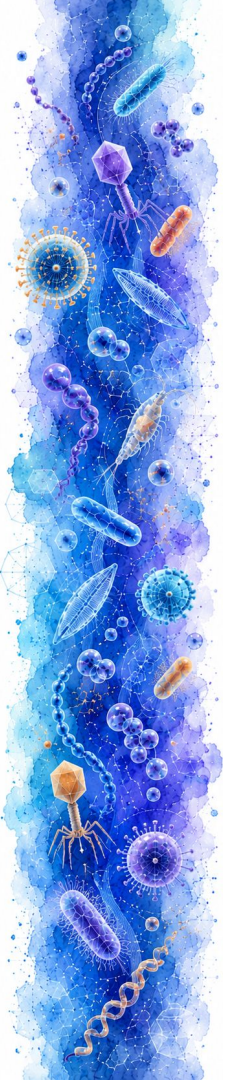
CONTACTS:

EMAIL: MATTEO.SELCI@UNINA.IT

TEAMS: XXX

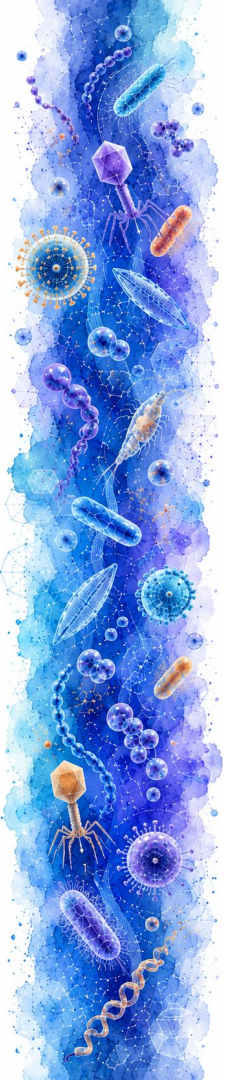
OFFICE HOURS: **TUESDAY**, FROM **2PM TO 4PM**, ROOM **1D10 - BIOLOGY DEPARTMENT**

(GET AN APPOINTMENT BY EMAIL AT LEAST **2 DAYS IN ADVANCE**)



Course Objectives

- .Getting an overview of the **diversity** and **role** of **microorganisms** in the **marine environment**
- .Getting an overview of the **techniques** and **approaches** used to study microbial diversity
- .Being able to recognize **major microbial groups** and their **metabolic and functional traits**
- .Being able to **ask relevant questions** regarding the role of microorganisms in the ecosystems
- .To have an idea of the **role, contribution** and **possible activities** of a microbiologist



HOW TO...

Ask! There's no such thing as a stupid question!

Search for papers using **DOI** (Digital Object Identifier)

Suggested books:

- >Brock Biology of Microorganisms 15th ed (Madigan, Bender, Buckley, Sattley, Stahl - Pearson Edition)
- >Marine Microbiology 3rd ed (Munn - Garland Science Edition)
- >Microbial Ecology of the Oceans 3rd ed (Gasol, Kirchman - Wiley Blackwell Edition)